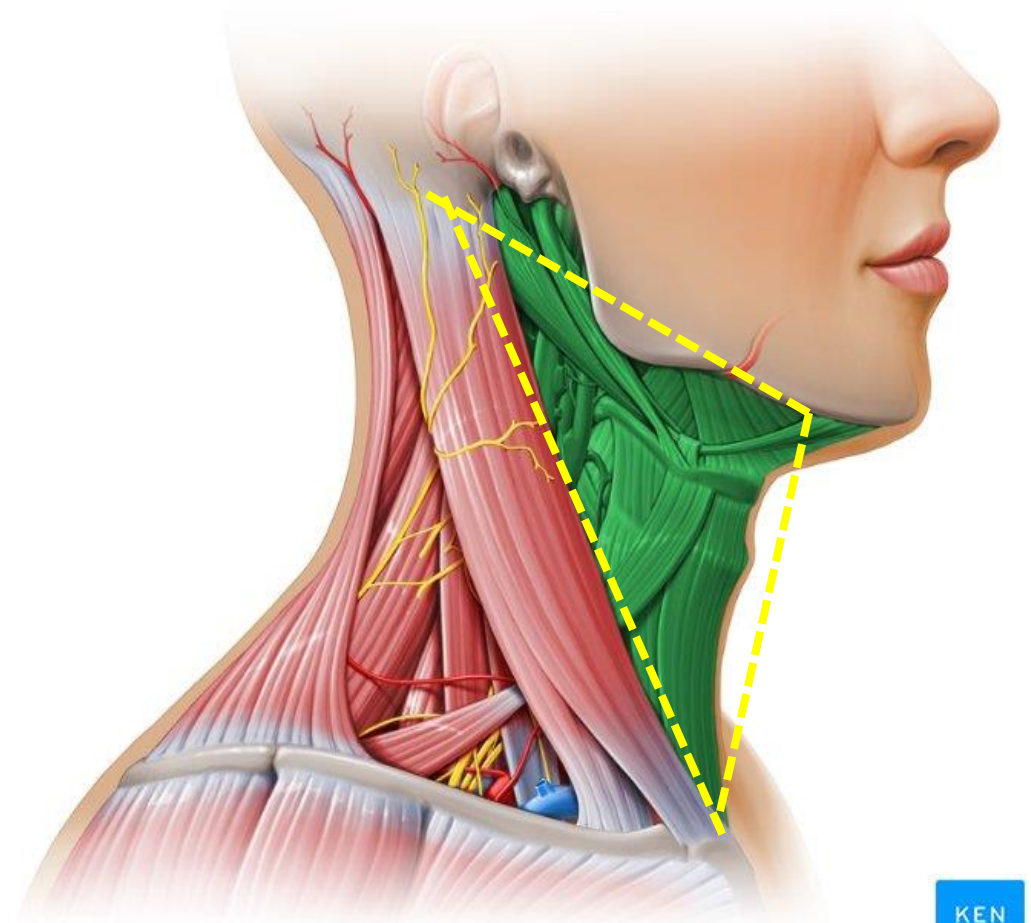


Anterior Triangle of the Neck

Boundaries:

Anterior triangle of the neck is bounded:

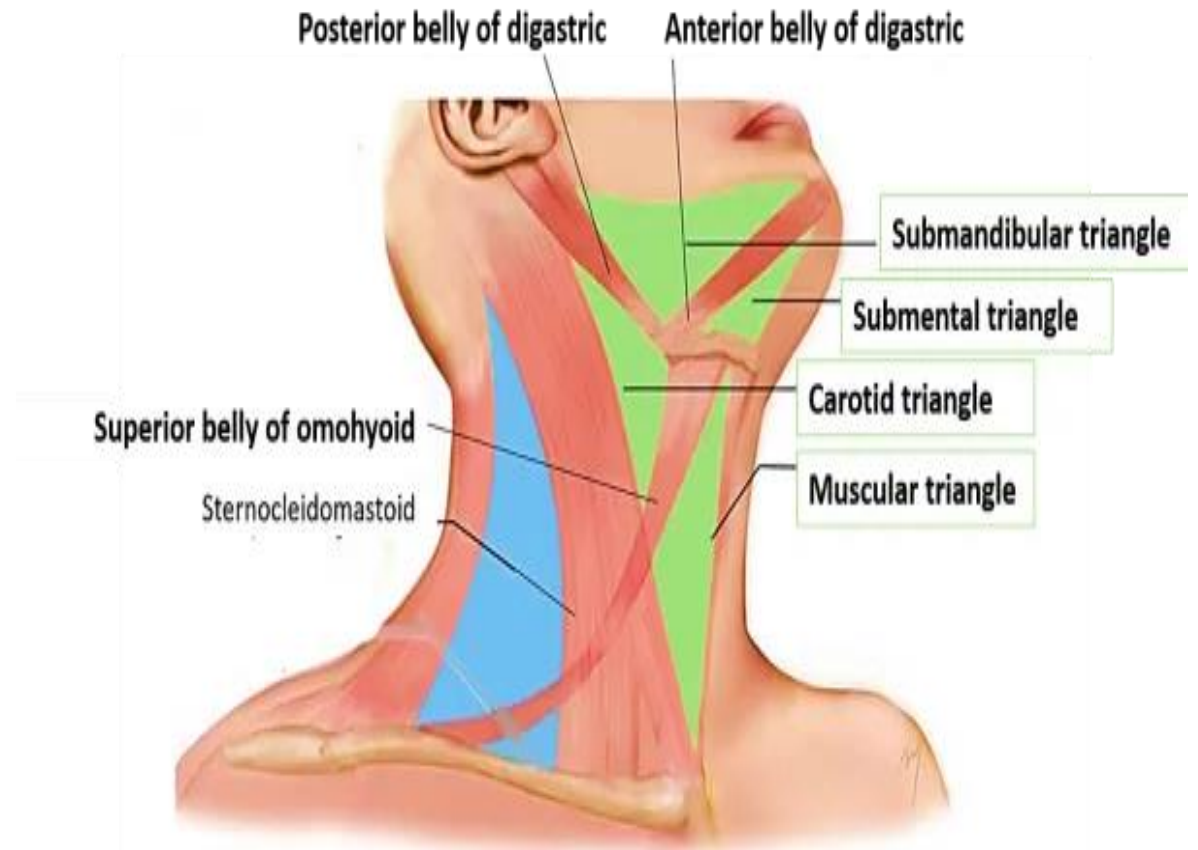
- **Anteriorly** by median plane
- **Posteriorly** by sternocleidomastoid muscle
- **Superiorly** by base of mandible and a line joining the angle of mandible to the mastoid process
- Apex lies at the manubrium sterni



Subdivisions:

The anterior triangle of the neck is subdivided by anterior and posterior bellies of the digastric muscle and superior belly of omohyoid into:

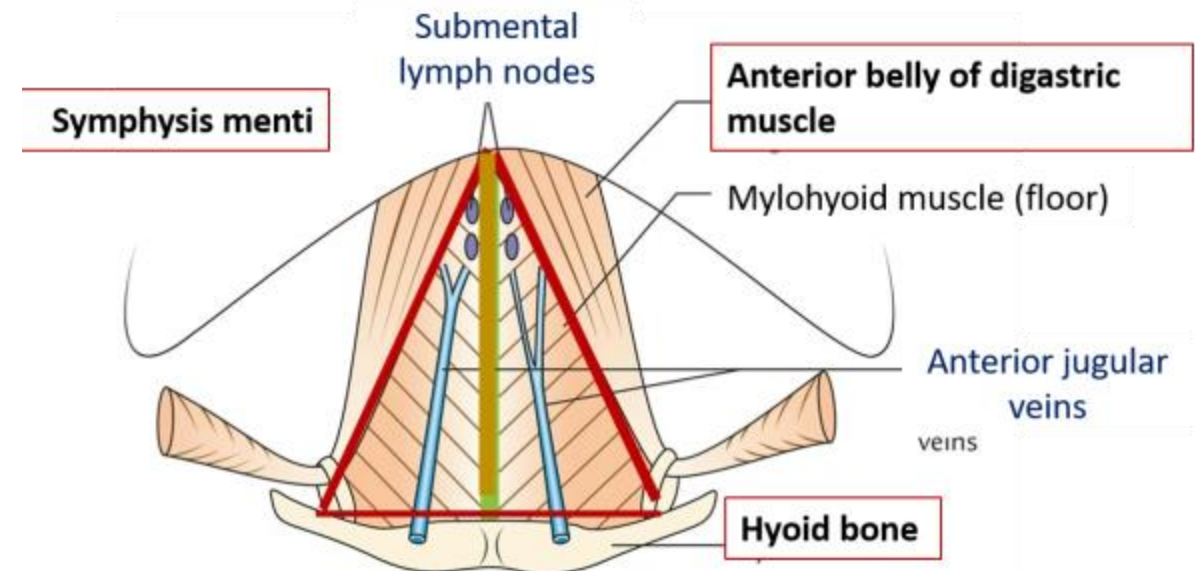
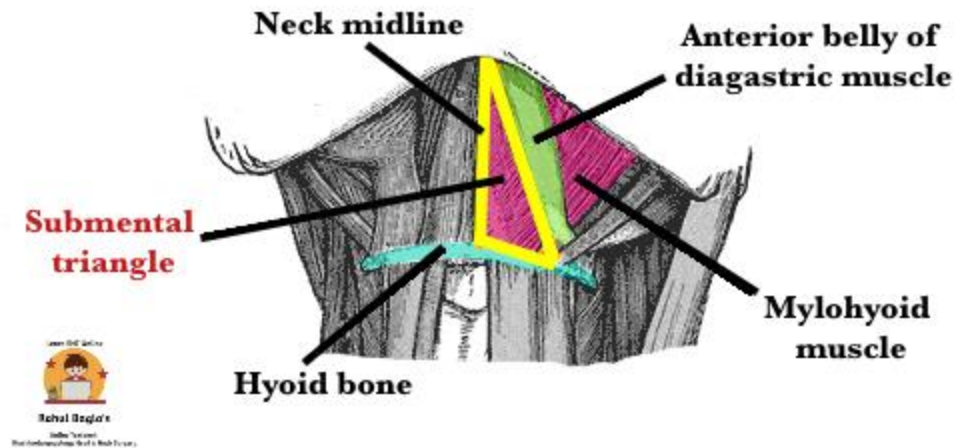
- 1) **Submental;**
- 2) **Digastric;**
- 3) **Carotid; &**
- 4) **Muscular triangles**



1) Submental Triangle:

It is a median triangle, which is bounded:

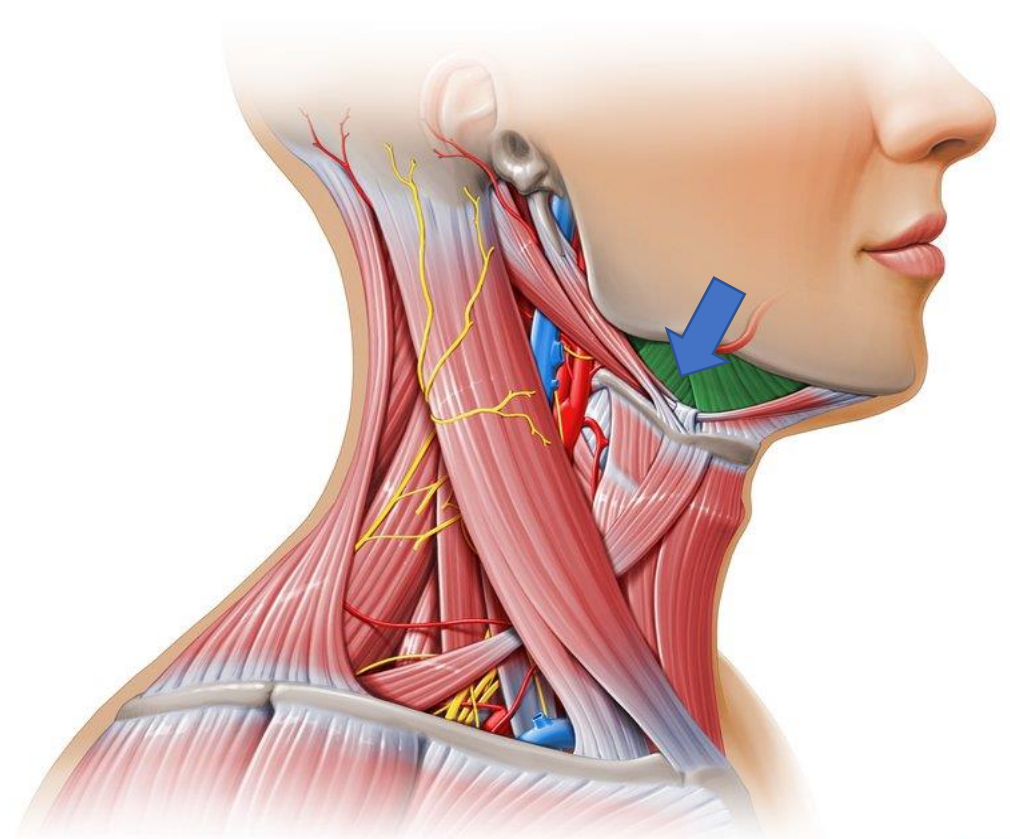
- **On each side** by anterior bellies of digastric muscles
- **Base** is formed by the body of hyoid bone
- **Floor** is formed by mylohyoid muscle
- **Contents:**
 - I. Submental lymph nodes
 - II. Submental veins unite to form anterior jugular vein



2) Digastric (submandibular) Triangle:

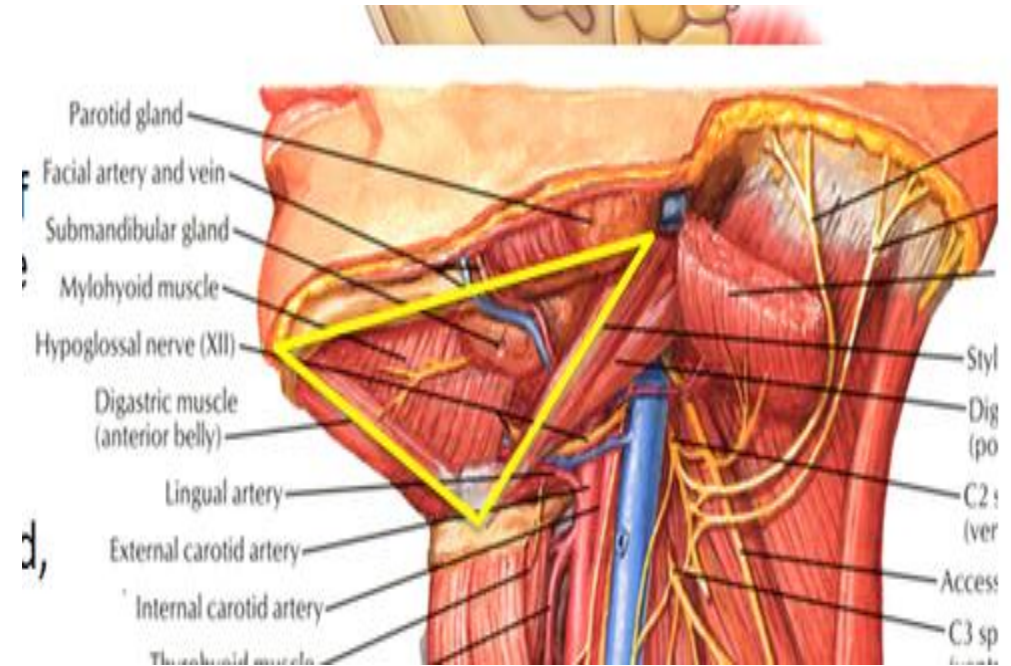
The boundaries of digastric triangle are:

- **Anteroinferiorly:** anterior belly of digastric
- **Posteriorly:** posterior belly of digastric & stylohyoid
- **Superiorly:** base of mandible and a line joining the angle of mandible to the mastoid process
- **Roof:** is formed by skin, superficial fascia containing platysma, cervical branch of facial nerve, and anterior cutaneous nerve of neck & investing layer of deep fascia
- **Floor:** is formed anteriorly by mylohyoid and posteriorly by hyoglossus muscle



- **Contents:**

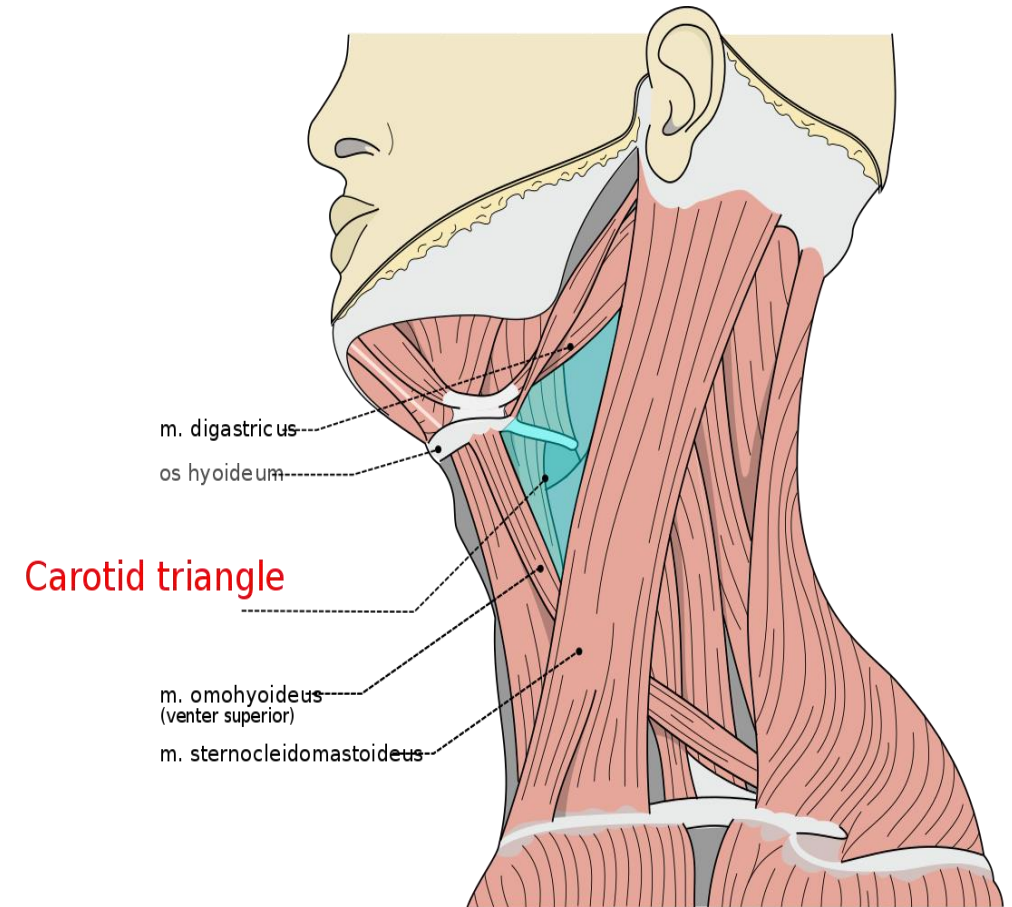
- I. Submandibular gland
- II. Facial vein (superficial to gland)
- III. Facial artery (deep to the gland) and its submental branch
- IV. Mylohyoid nerve & vessels
- V. Hypoglossal nerve
- VI. Lower part of parotid gland
- VII. Part of external carotid artery
- VIII. Glossopharyngeal nerve



3) Carotid Triangle:

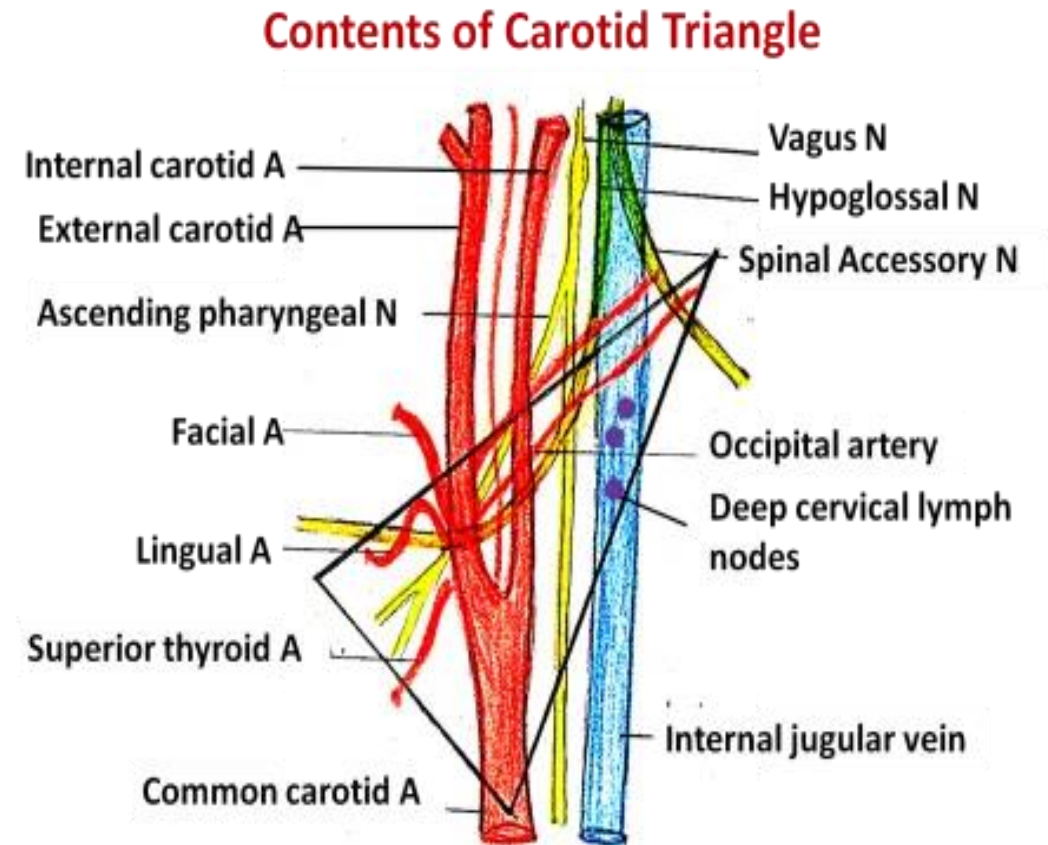
The boundaries of carotid triangle are:

- **Anterosuperiorly:** posterior belly of digastric and stylohyoid muscle
- **Anteroinferiorly:** by superior belly of omohyoid
- **Posteriorly:** anterior border of sternocleidomastoid muscle
- **Roof:** is formed by skin, superficial fascia containing platysma and anterior cutaneous nerve of neck & investing layer of deep cervical fascia
- **Floor:** is formed by contributions from thyrohyoid membrane, middle and inferior constrictors of pharynx



- **Contents:**

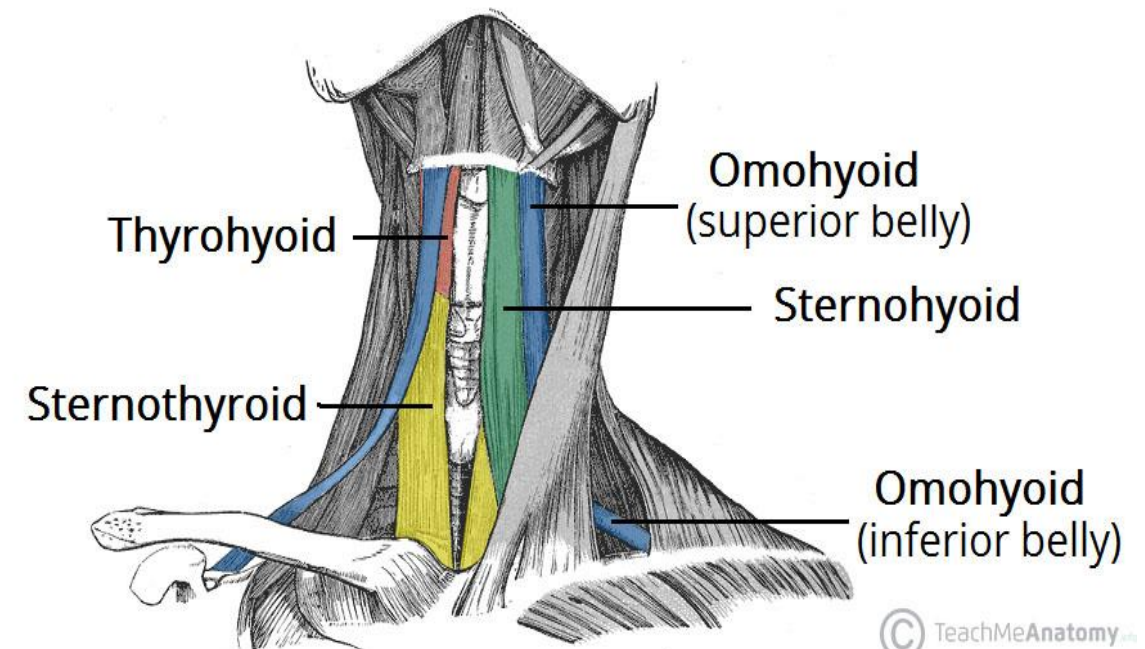
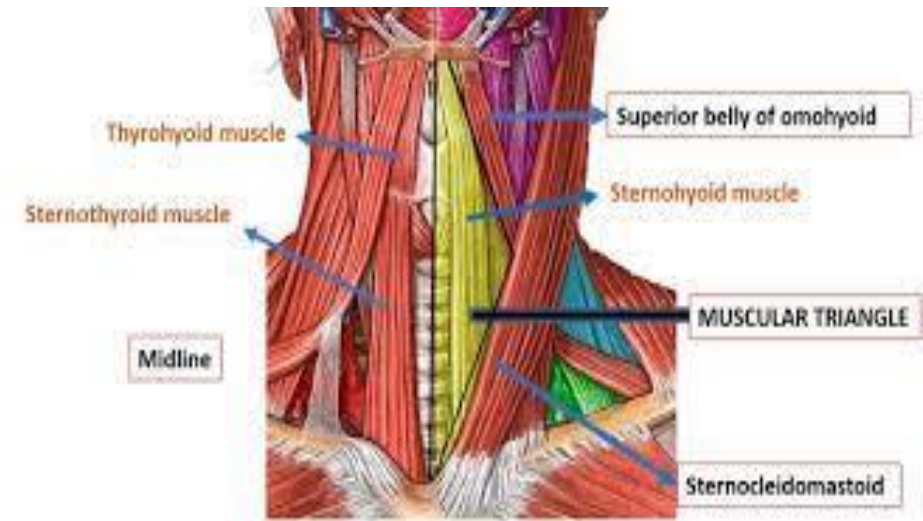
- I. **Arteries:** common carotid artery, internal carotid artery and external carotid artery with its superior thyroid, lingual, facial, ascending pharyngeal and occipital branches
- II. **Veins:** internal jugular vein, common facial vein, pharyngeal vein and lingual vein
- III. **Nerves:** vagus nerve, superior laryngeal branch of vagus dividing into external and internal laryngeal nerves, spinal accessory nerve, hypoglossal nerve and sympathetic chain
- IV. **Carotid sheath with contents**
- V. Deep cervical (jugulodigastric and juguloomohyoid) lymph nodes



4) Muscular Triangle:

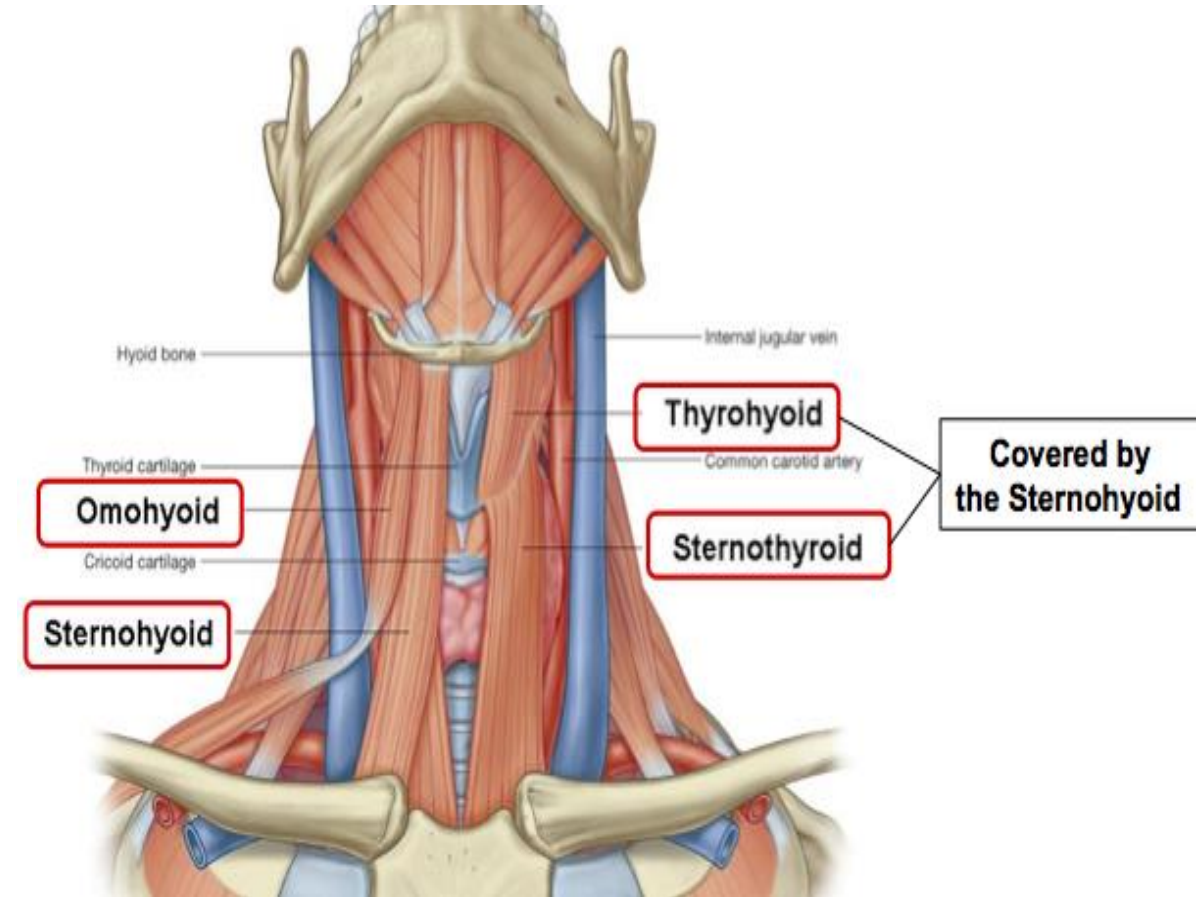
The boundaries of muscular triangle are as follows:

- Anteriorly: midline of the neck from hyoid bone to sternum
- Posterosuperiorly: superior belly of omohyoid muscle
- Posteroinferiorly: anterior border of sternocleidomastoid muscle
- **Contents:** chief contents are infrahyoid muscles (sternohyoid, sternothyroid, thyrohyoid & omohyoid).
- They also form the floor of muscular triangle



I. Infrahyoid Muscles

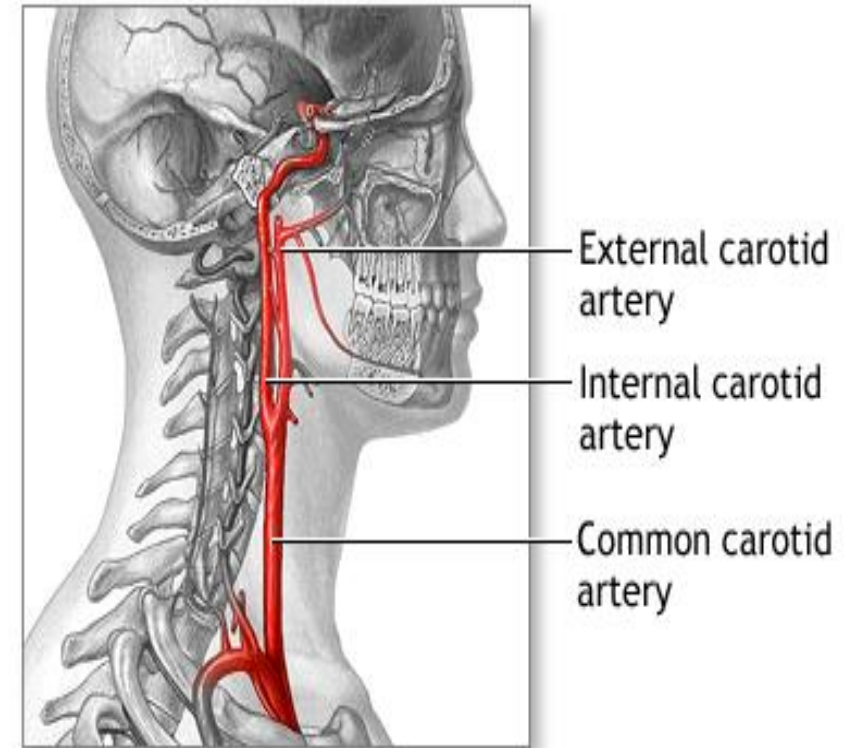
Muscle	Origin	Insertion	Action	Nerve Supply
Omohyoid	<u>Inferior belly:</u> Superior border of the scapula <u>Superior belly:</u> Intermediate tendon	<u>Inferior belly:</u> Intermediate tendon <u>Superior belly:</u> Lower border of the body of the hyoid bone	Depresses the hyoid bone and larynx	Ansa cervicalis (APR of C1, C2, C3)
Sternohyoid	Manubrium of the sternum, posterior aspect	Lower border of the body of the hyoid bone	Depresses the hyoid bone and larynx	Ansa cervicalis (APR of C1, C2, C3)



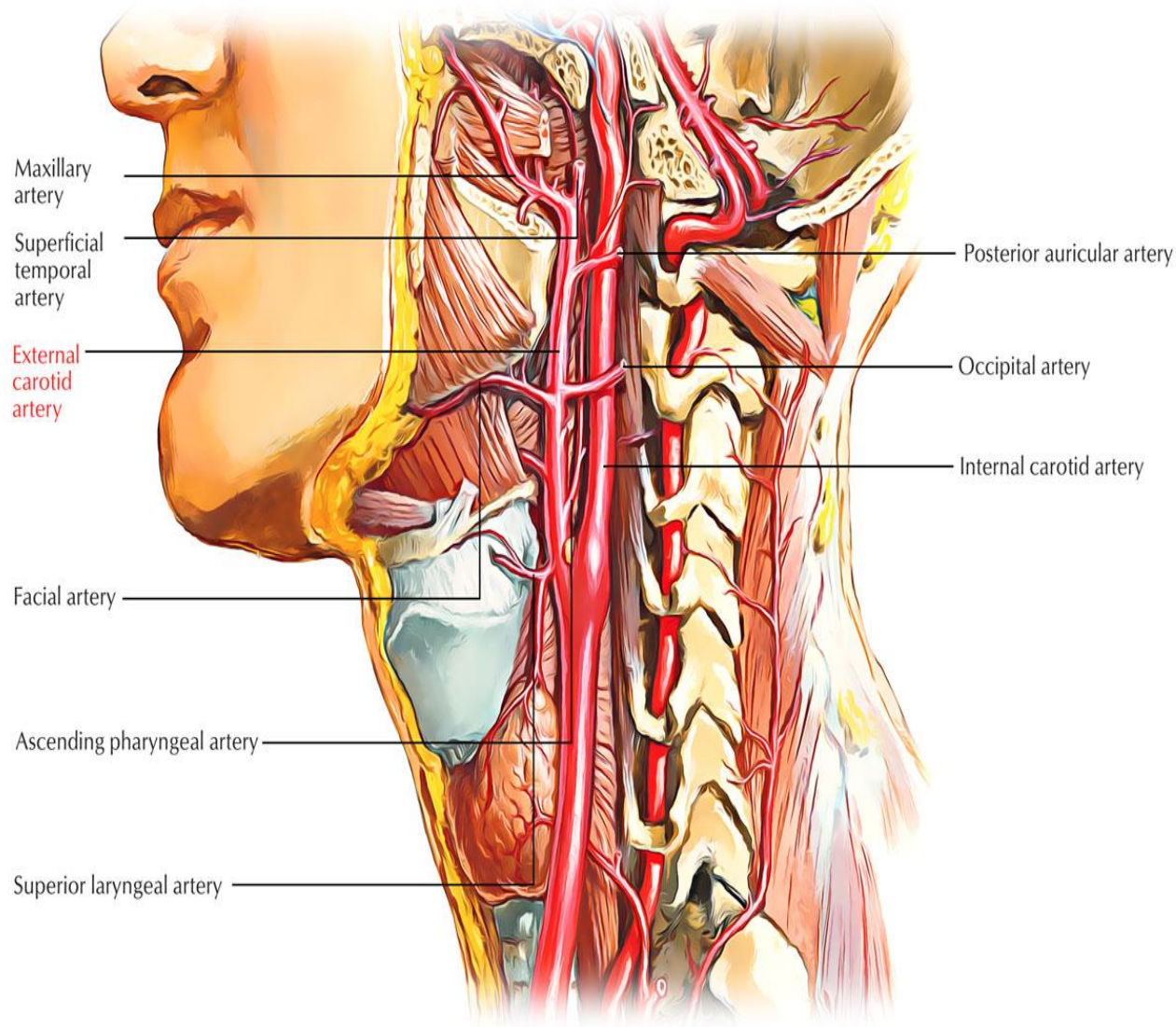
Muscle	Origin	Insertion	Action	Nerve Supply
Sternothyroid	Manubrium of the sternum, posterior aspect	Oblique line of thyroid cartilage	Depresses the larynx	Ansa cervicalis (APR of C1, C2, C3)
Thyrohyoid	Oblique line of thyroid cartilage	Body and greater horn of hyoid bone, lower border	Depresses the hyoid bone	Thyrohyoid branch of the hypoglossal nerve

The Carotid arteries:

- **COMMON CAROTID ARTERY**
- **Origin:**
- **Right-** from the **brachiocephalic trunk** (Sterno clavicular joint)
- **Left-** **Arch of aorta** (Manubrium sterni)
- **Termination:**
- **Upper border of lamina of thyroid cartilage** divides into **external and internal carotid.**
 - carotid triangle (C3 & C4 vertebra).



External Carotid artery:



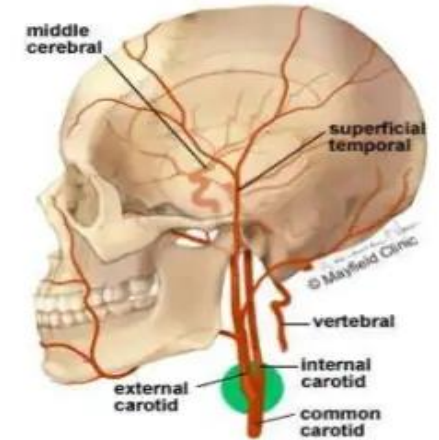
EXTERNAL CAROTID ARTERY

It arises from the common carotid artery when it bifurcates into the external and internal carotid artery.

Course:

- It begins at the level of the upper border of thyroid cartilage.

- It passes upward and forward, and then inclines backward to the space behind the neck of the mandible, where it divides into the superficial temporal and maxillary artery within the parotid gland.



External carotid artery branch	Position/location on the external carotid artery	Structure it supplies	Other features
Superior thyroid artery	First branch – anterior surface	Thyroid gland	Arises below greater cornu of the hyoid bone
Lingual artery	Second branch – anterior surface	Tongue	Arises opposite greater cornu of the hyoid bone
Facial artery	Third branch – anterior surface	Face and submandibular gland	Deep to the hypoglossal nerve at its origin
Ascending pharyngeal artery	Medial surface, close to the carotid bifurcation	Constrictor muscles of the pharynx	Ascends vertically between the internal carotid artery and pharynx
Occipital artery	Posterior surface	Sternocleidomastoid muscle	Crossed by the hypoglossal nerve at its origin
Posterior auricular artery	Posterolateral surface	Digastric muscle	Arises above the digastric and stylohyoid muscles
Superficial temporal artery	Lateral terminal branch of the external carotid artery	Temporalis muscle	Arises in the parotid gland
Maxillary artery	Medial/ deep terminal branch of the external carotid artery	Buccinator muscle	Third part located in the pterygopalatine fossa of the maxilla

Applied anatomy

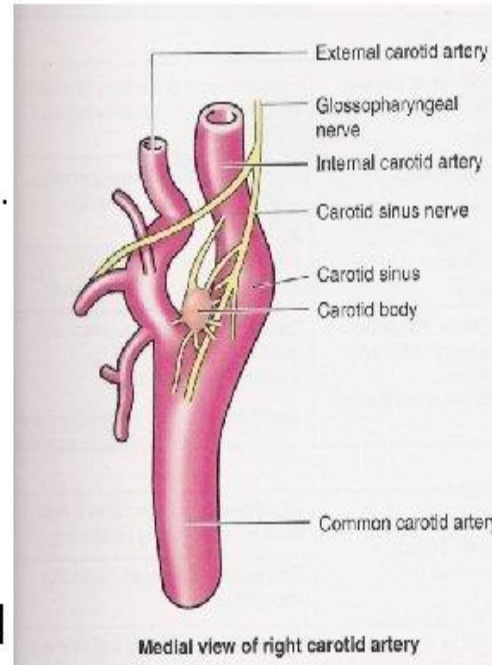
⌘ Carotid sinus

⌘ Dilatation –termination of CCA/beginning of ICA.

⌘ Glossopharyngeal & symp nerves.

⌘ **Baroreceptor**/pressure receptor.

⌘ **Carotid sinus syndrome**: syncope and slowing of heart rate.



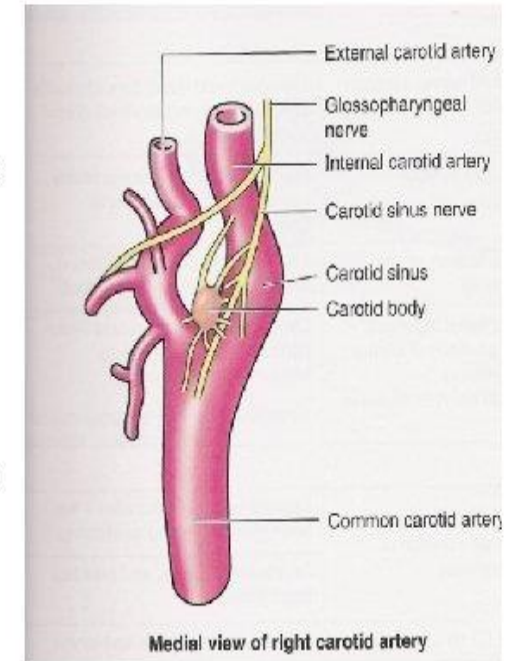
⌘ Carotid body

⌘ Oval, reddish mass-bifurcation of CCA.

⌘ Supplied by glossopharyngeal, vagus & symp nerves.

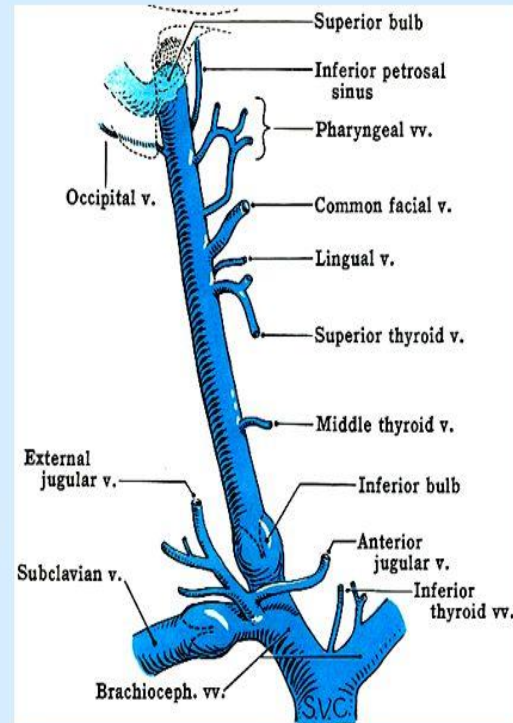
⌘ **Chemoreceptor**, responds to changes O₂ & CO₂ changes in blood.

⌘ **Potato tumor**: enlargement of the carotid body. Pressure on carotid sinus.



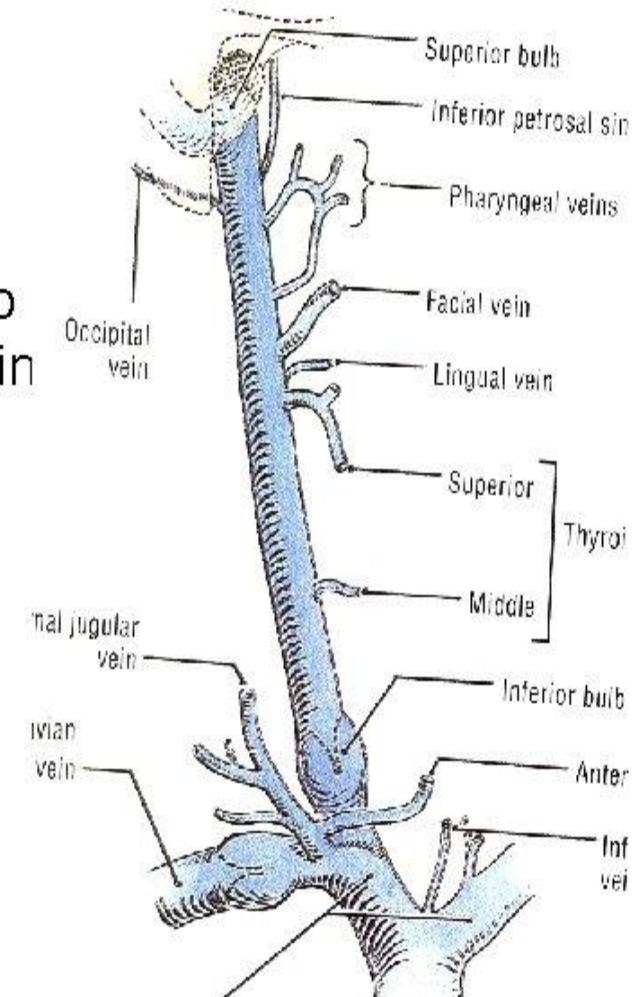
Internal jugular vein:

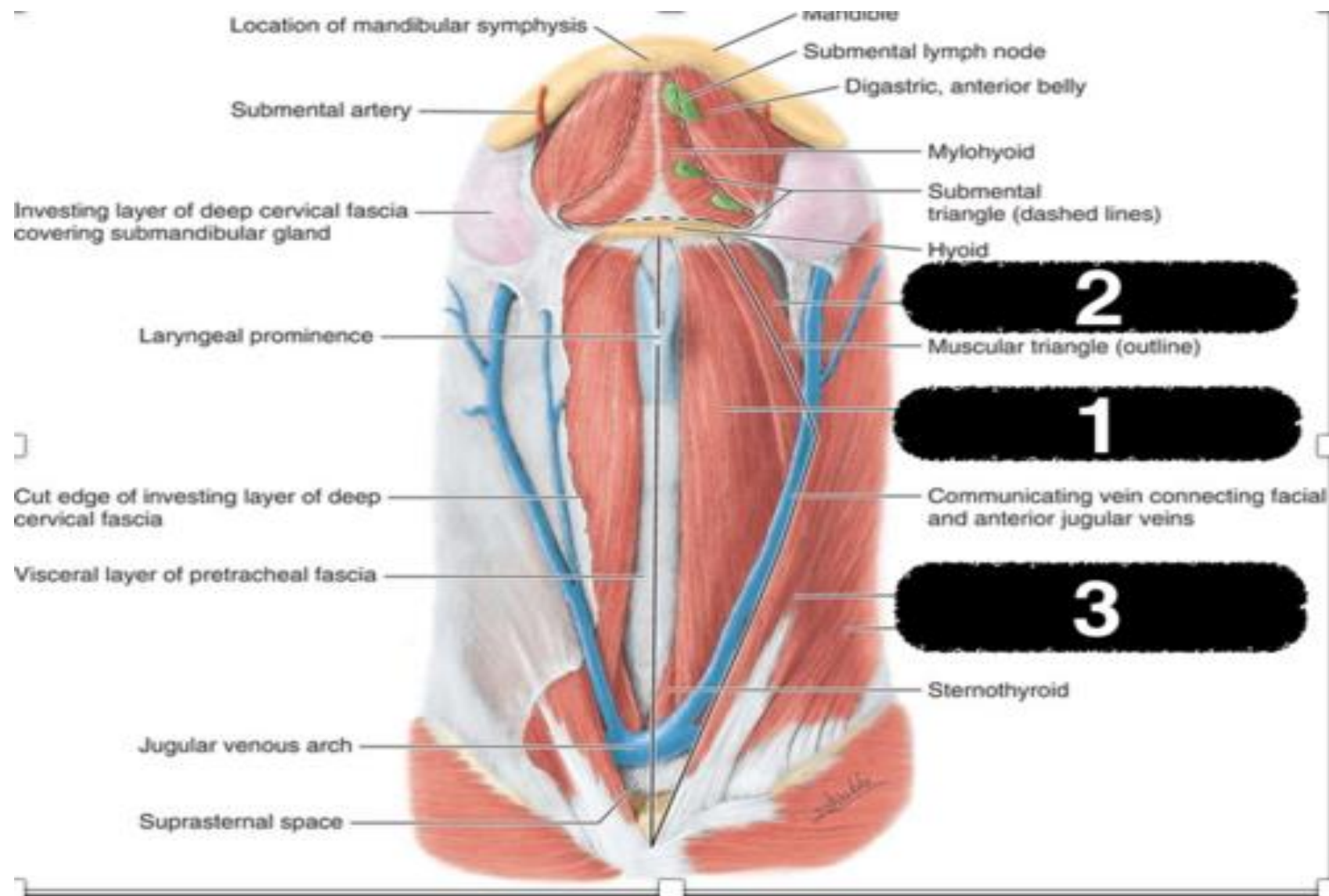
- It receives blood from the brain, face and the neck.
- It emerges through the jugular foramen, as a continuation of the sigmoid sinus descend down in the neck, first behind then lateral to the internal carotid artery inside the carotid sheath



VEINS:

- Internal jugular veins.
- Five tributaries to internal jugular vein
 - pharyngeal veins
 - common facial,
 - lingual,
 - superior thyroid,
 - middle thyroid,





thank you!